



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 1 – Fundamentals of OOP
Topic –1.1 Introduction to procedural, modular, object-oriented and generic programming techniques

Prof. V.N.Nirgude

Assistant Professor

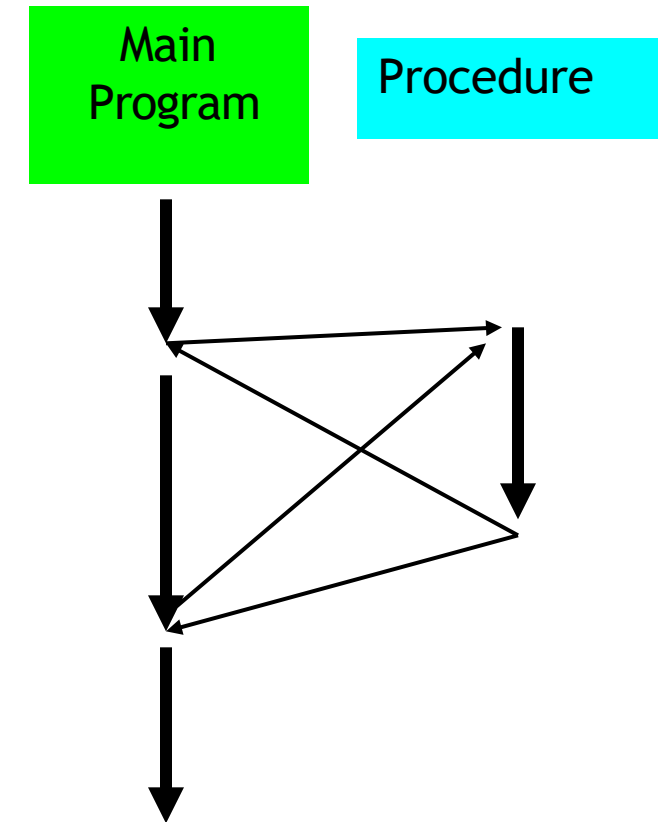
E-mail :

nirgudevikascomp@sanjivani.org.in

Contact No: 9975240215

Procedural Programming

- With procedural programming, you are able to combine sequences of calling statements into one single place.
- A procedure call is used to invoke the procedure. After the sequence is processed, flow of control proceeds right after the position where the call was made.





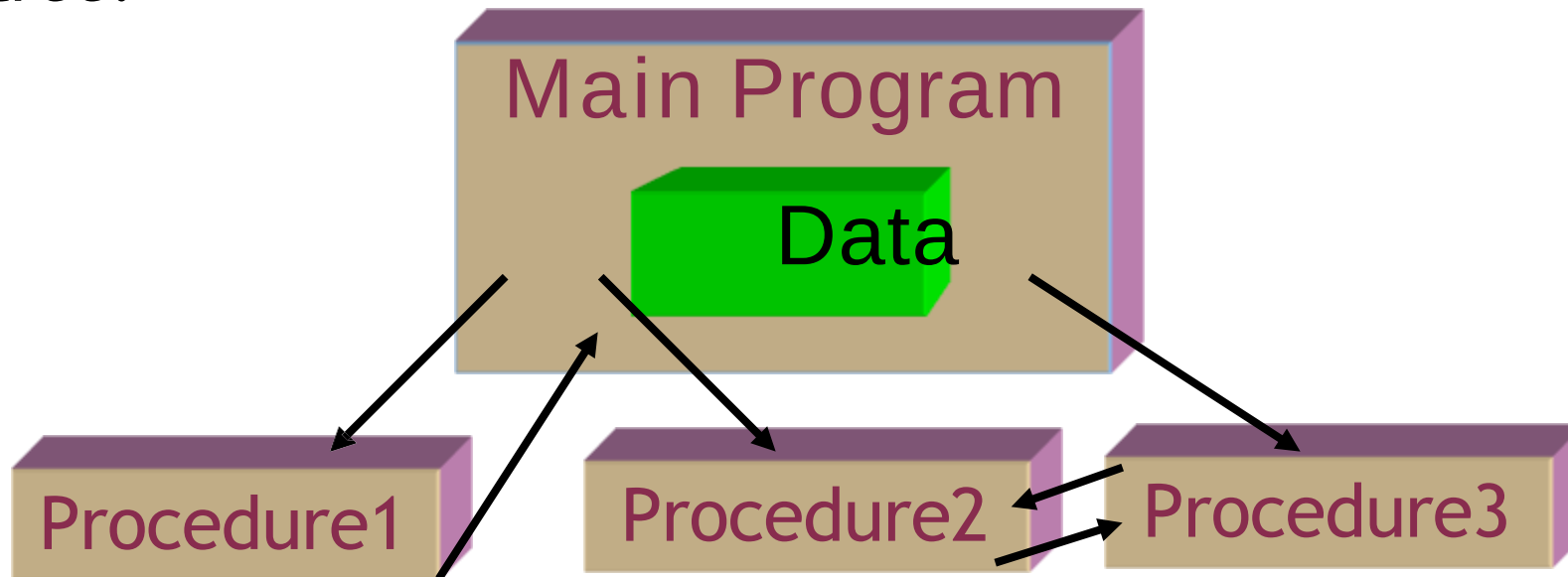
Procedures

- Using procedures programs can be written more structured and error free.
- For example, if a procedure is correct, every time it is used it produces correct results.
- Consequently, in cases of errors you can narrow your search to those places which are not proven to be correct.



Procedure Program View

- Now a program can be viewed as a sequence of procedure calls.
- The main program is responsible to pass data to the individual calls, the data is processed by the procedures and the resulting data is presented back to main program.
- Thus, the flow of data can be illustrated as a hierarchical graph or a tree.



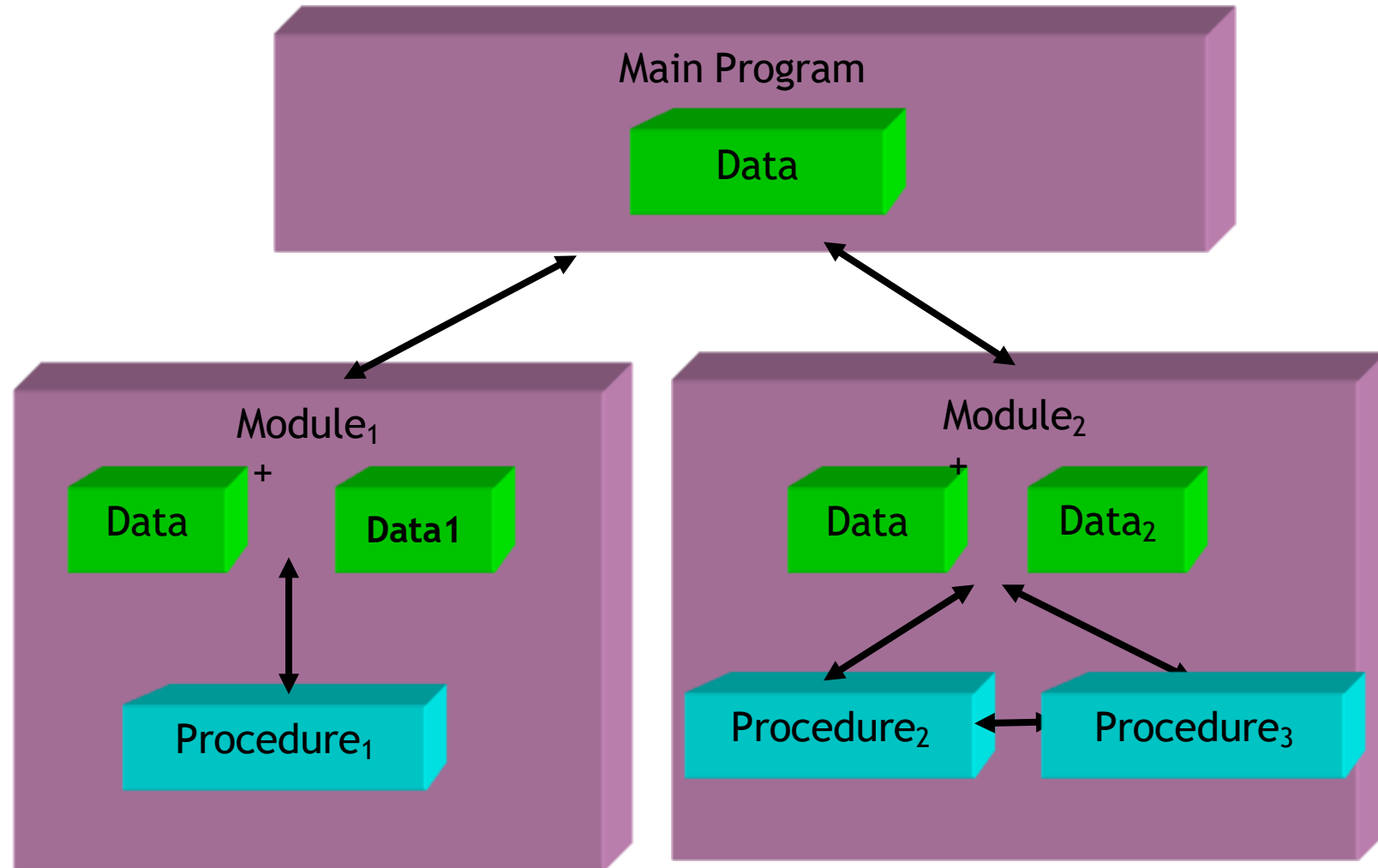


Modular Programming

- Modular programming is subdividing your program into separate subprograms such as functions and subroutines.
- With modular programming, procedures of a common functionality are grouped together into separate modules.
- A program therefore no longer consists of only one single part. It is now divided into several smaller parts which interact through procedure calls and which form the whole program.

Modular Programming

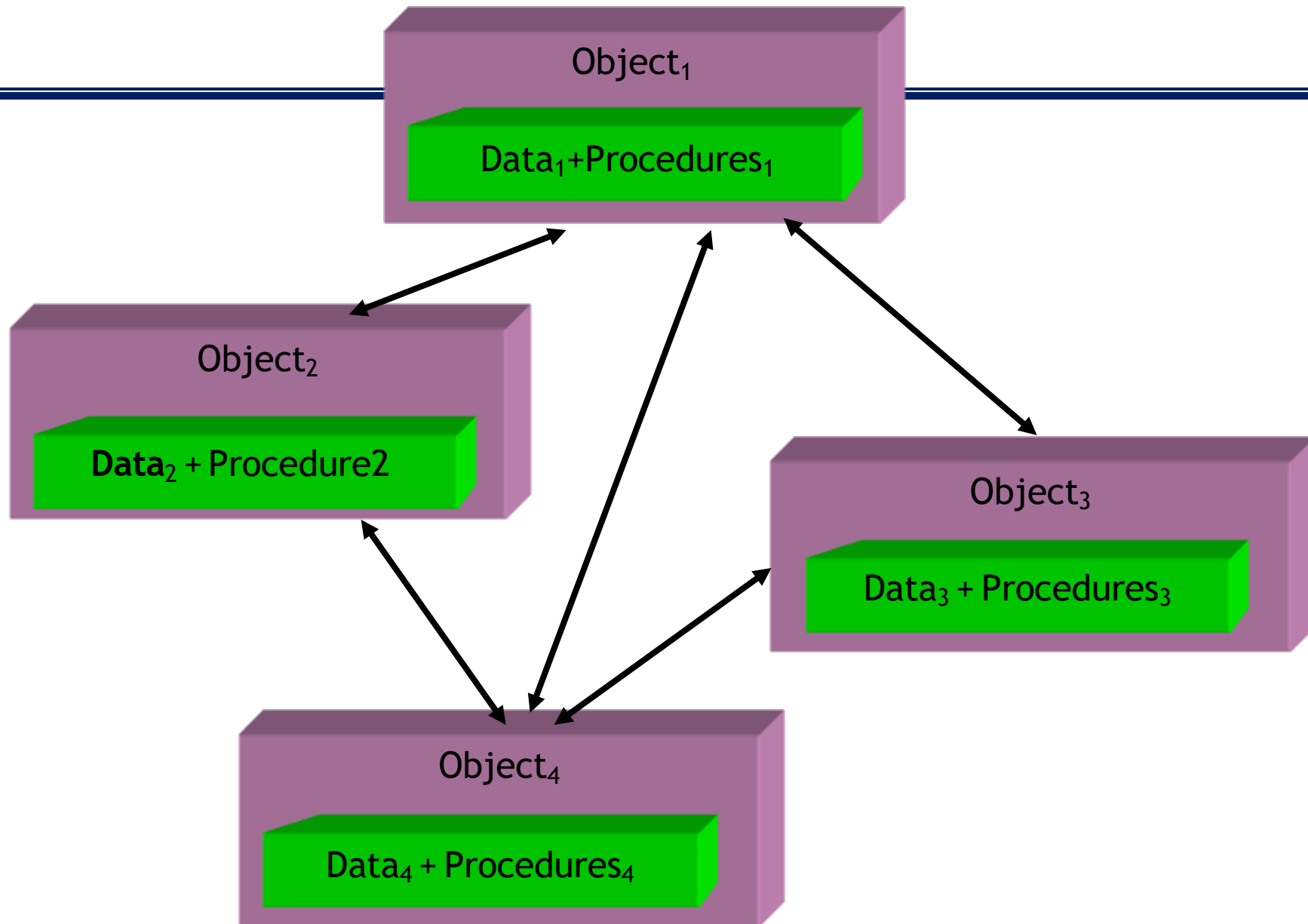
- Each module can have its own data. This allows each module to manage an internal state which is modified by calls to procedures of this module.
- Each module has its own special functionalities that supports the implementation of the whole program.





Object Oriented Programming

- Build programming using software objects.
- Means that we organize software as a collection of discrete (different) objects that incorporate both data structure and behavior.
- In OOP, an object correspond closely to real objects involve in the application area.
- Everything in the world is an object..like a flower, a tree, an animal, a student, a chair, a building, a city, a country, a subject such as CS, Math, History etc.





Object Oriented Languages

- An object-based programming language is one which easily supports object-orientation.
- Smalltalk: 1972-1980, Alan Kay
- C++: 1986, Bjarne Stroustrup
- Java: 1992 (Smalltalk + C++), James Gosling
- C#: Developed at Microsoft by Anders Hejlsberg, 2000



Generic programming

- Generic programming is a style of computer programming in which algorithms are written in terms of types to-be-specified-later that are instantiated when needed for specific types provided as parameters.
- Generic programming makes developers' life easier by giving us the ability to write code once and reuse it across different types in a typesafe manner. The majority of strongly typed languages support generic programming.



Generic Code illustration with

Example

Code for swapping of integer values:

```
void Swap(int a, int b)
{
    int temp = b;
    b = a;
    a = temp;
}
```

Generic Code for swapping of any type of values:

```
template <typename T>
void Swap(T& a, T& b)
{
    T temp = b;
    b = a;
    a = temp;
}
```

Thank you...